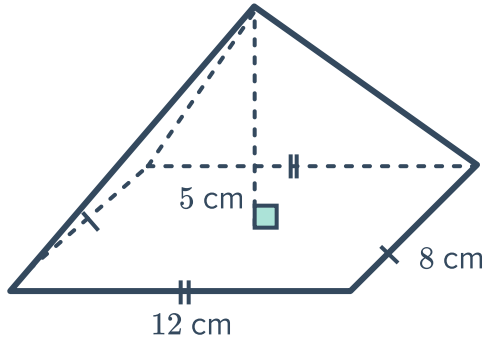


Volume of pyramids and cones

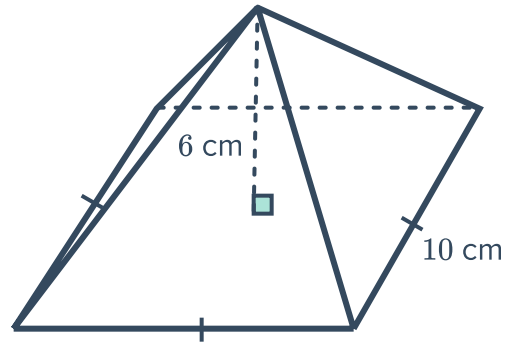


1 Find the volume of the following pyramids, correct to two decimal places if necessary:

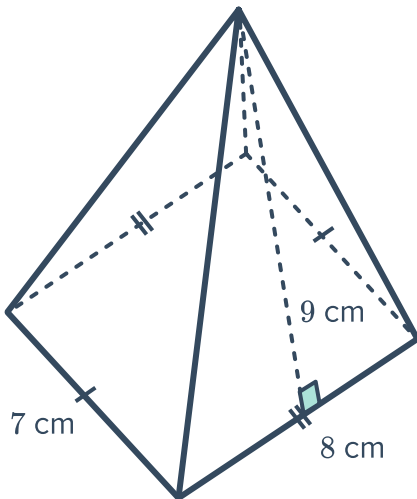
a



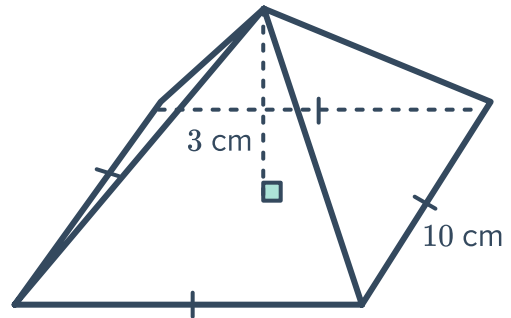
b



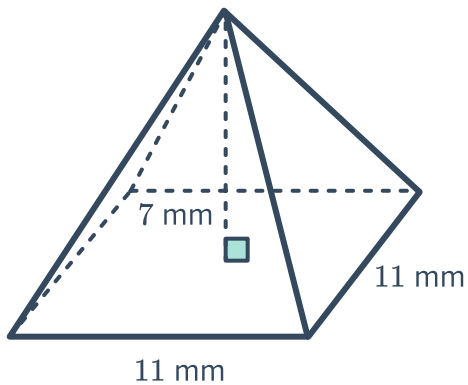
c



d

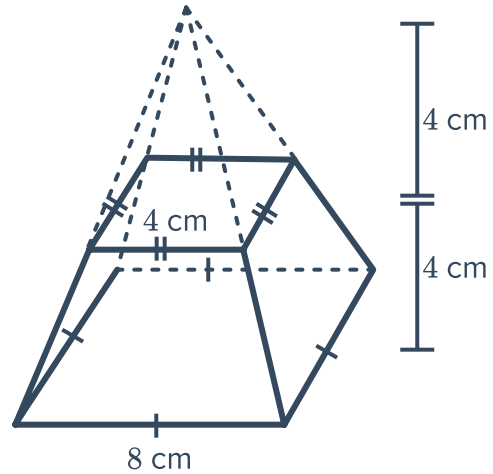


e



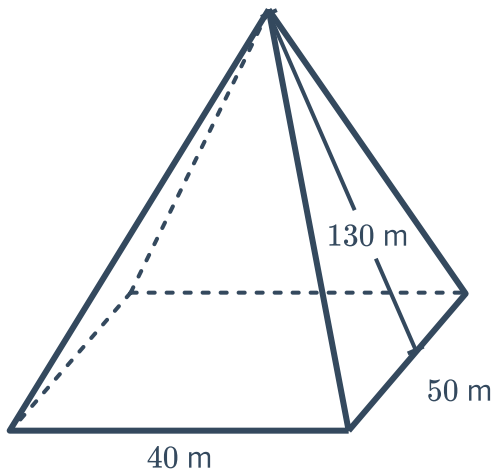


- 2 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown. Find the exact volume of the solid.

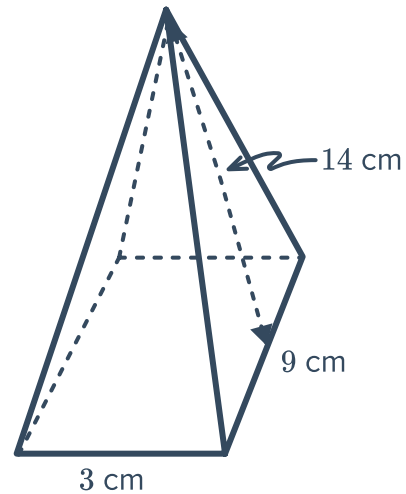


- 3 Suppose one particular pyramid in Egypt has a square base of side length 231 m and with a perpendicular height of 144 m. Find the volume of the pyramid, correct to the nearest m^3 .
- 4 A rectangular pyramid has a volume of 288 cm^3 . The base has a width of 12 cm and length 6 cm. Find the height of the pyramid.
- 5 Consider the following right pyramids:
- a Find the perpendicular height of the pyramid, correct to two decimal places.
- b Hence, find the volume of the pyramid, correct to two decimal places.

a



b

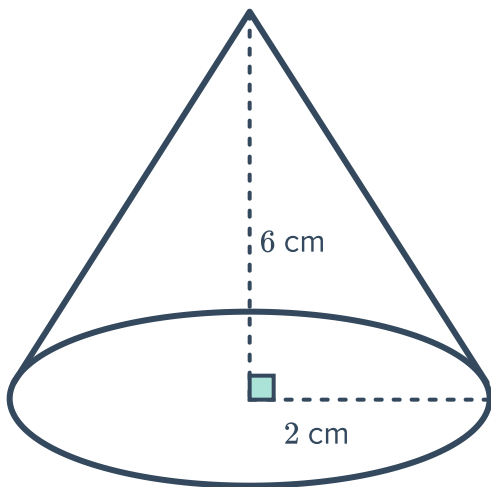


- 6 Some very famous right pyramids are the Egyptian Pyramids.
The diagram shows the Great Pyramid. It has a square base of side length 230 m and a vertical height of 146 m. Find the volume of the Great Pyramid, correct to the nearest tenth.

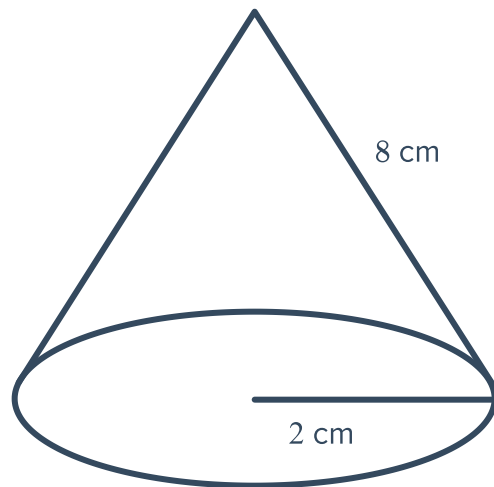


- 7 Find the volume of the following right pyramids:
- a A right pyramid with a square base of 24 mm and a perpendicular height of 24 mm.
 - b A right pyramid with a square base of 21 mm and a vertical height of 27 mm.
- 8 A right square pyramid has a base of length 12 mm and a volume 1248 mm^3 . Find the height of the pyramid.
- 9 A right square pyramid has a height of 24 cm and a volume 2592 cm^3 . Find the base length of the pyramid.
- 10 Find the volume of the following cones, correct to two decimal places:

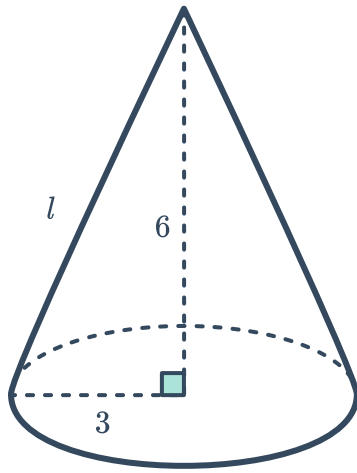
a



b

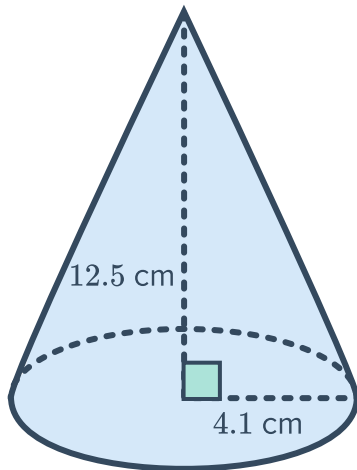


c

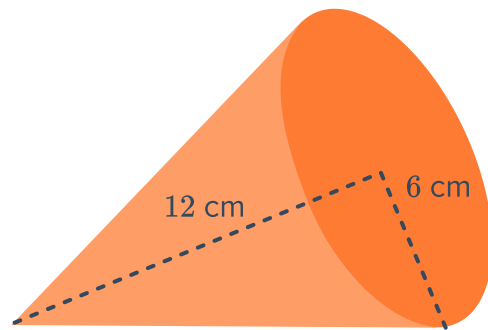


11 Find the volume of the following cones, correct to one decimal place:

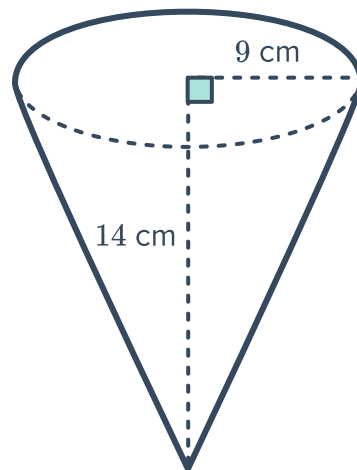
a



b



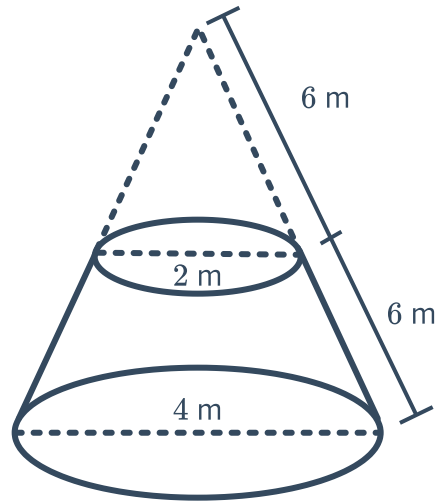
c



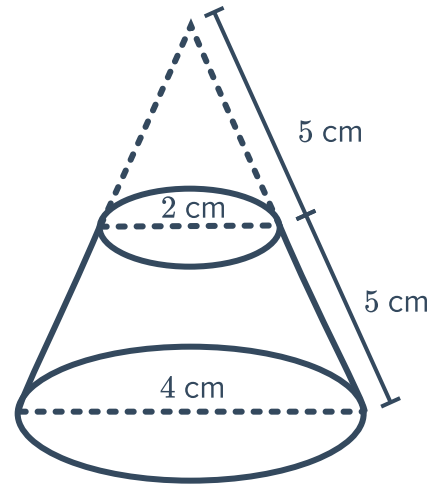
- 12 The podium was formed by sawing off the top of a cone. Find the volume of the following podiums, correct to two decimal places:



a



b



- 13 Find the volume of the following cones, correct to two decimal places:

- a A cone with radius of 8 cm and vertical height of 13 cm.
- b A cone with radius of 60 m and a vertical height of 27.54 m.

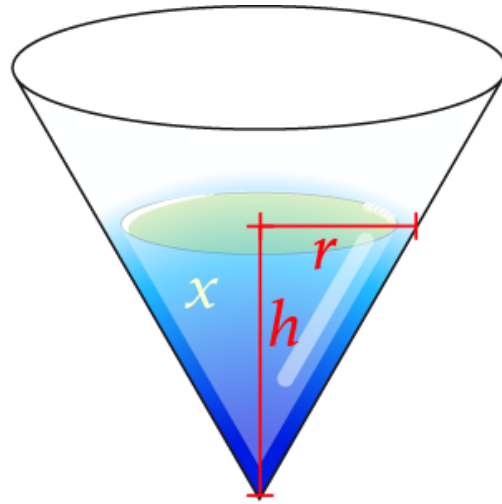
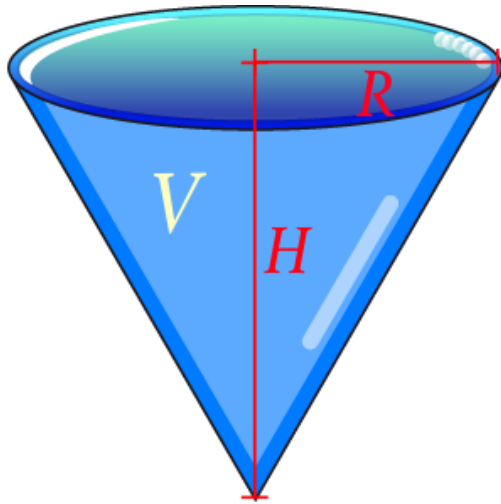
- 14 Find the height of the following cones:

- a A cone with volume 603.19 m^3 and radius 6 m. Round your answer to one decimal place.
- b A cone with volume $26\,640.97 \text{ cm}^3$ and radius 31.9 cm. Round your answer to the nearest cm.

- 15 Find the radius of the following cones:

- a A cone with volume of $12\,441.02 \text{ cm}^3$ and a height of 30 cm. Round your answer to one decimal place.
- b A cone that has volume 1273.39 mm^3 and height 19 mm. Round your answer to the nearest mm.
- c A cone with volume of 196 mm^3 and the length of the height and radius are equal. Round your answer to two decimal places.

- 16 A water vessel is produced in the shape of a cone of radius R and maximum volume V . From vertex to base, the vessel has a height of H . A volume x of water is poured into the vessel, filling up from the vertex towards the base. The radius of the cone of water within the vessel has radius r and height h .

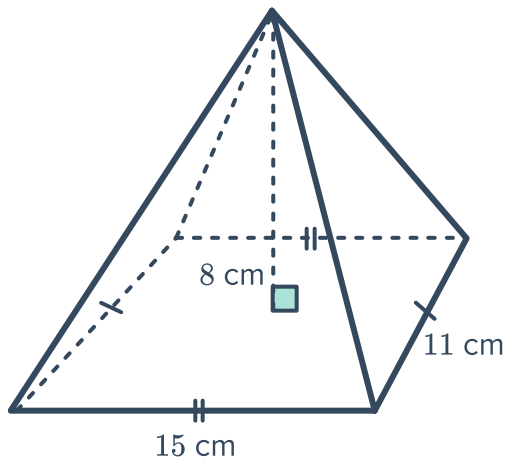


Write an expression for the height of the water, h , in terms of the height of the vessel H and the volumes V and x .

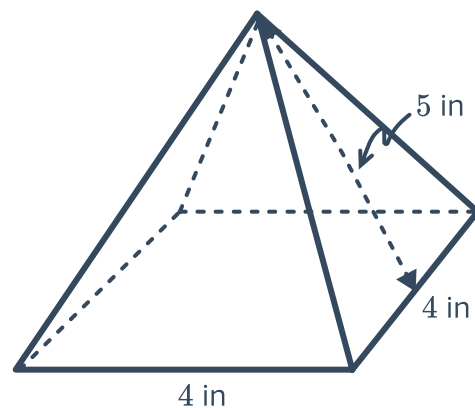
Additional questions

- 17 Find the volume of the following solids, rounding to two decimal places if necessary:

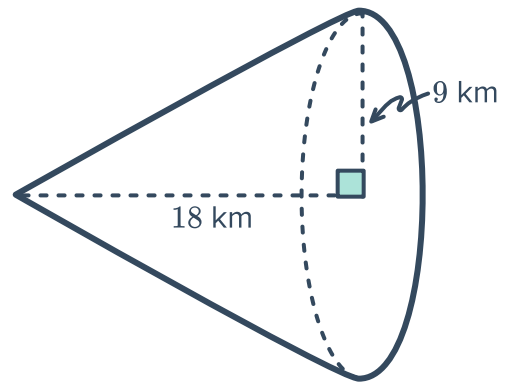
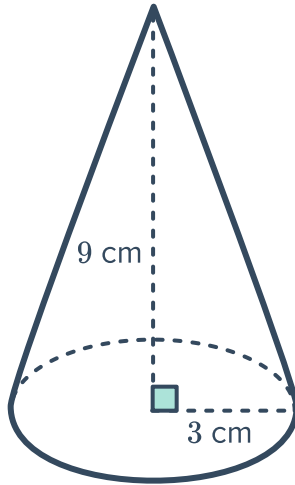
a



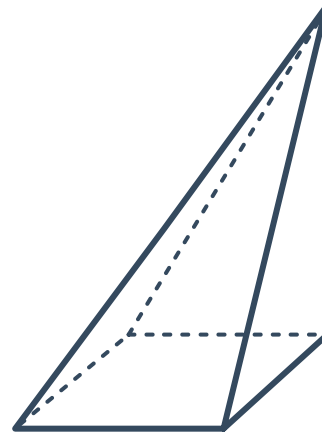
b



c



- 18 The volume of a right square pyramid is 55 in^2 . If the pyramid is slanted by 60° , find its volume.



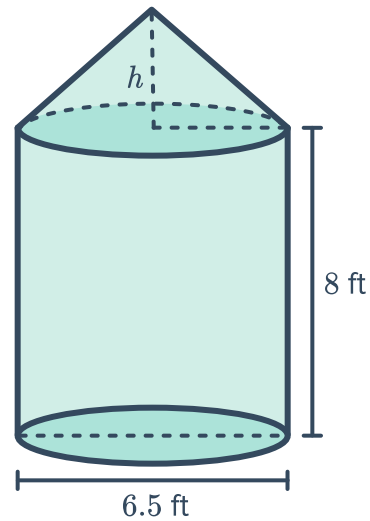
- 19 The radius and the height of a cone are equal, and the volume of the cone is 72π . Find the radius of the cone.

- 20 The volume of a pyramid is 60 m^3 with a height of 4 m . Find the area of the base.

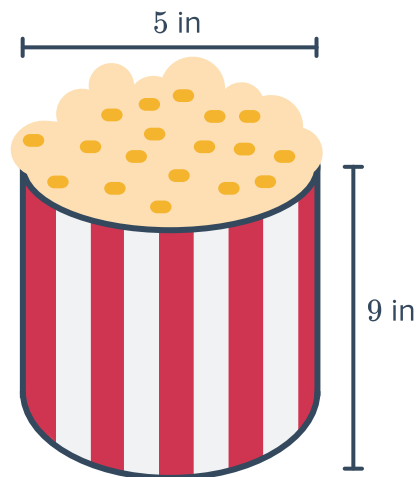
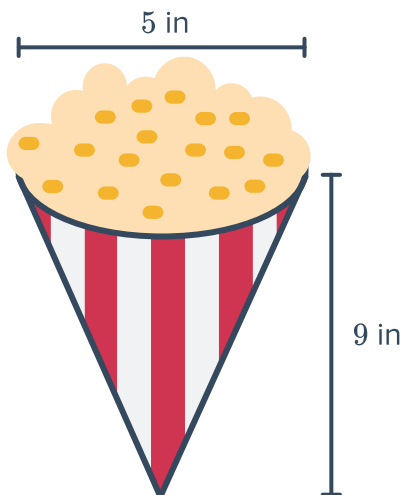
- 21 The Florida Agricultural Museum plans to bring in a replica grain silo for a new exhibit. The volume of the conical portion of the silo is $\frac{169}{24}\pi \text{ ft}^3$.



The museum's doorway has a height clearance of 12 ft. Will the model grain silo fit through the door? Explain.



- 22 A vendor at the Sumter County Fair sells kettlecorn in two containers.
Option 1 sells for \$3.50: Option 2 sells for \$10.50:



If you have \$12.00 to spend, what would you order to get the maximum amount of kettlecorn? Explain.